Shuai Mu

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Professional Experience

STONY BROOK UNIVERSITY, COMPUTER SCIENCE DEPARTMENT — Stony Brook, NY

2025–Present Associate Professor

2018–2025 Assistant Professor; initial appointment August 2018, on leave until January 2019

NEW YORK UNIVERSITY, COURANT INSTITUTE — New York, NY

2017/08–2018/08 Assistant Professor / Faculty Fellow (Lecturer)

2015/08–2017/08 Post-Doctoral Associate; advisor: Michael Walfish

UNIVERSITY OF SOUTHERN CALIFORNIA — Los Angeles, CA

2015/01–2015/07 Visiting Scholar; advisor: Wyatt Lloyd

NEW YORK UNIVERSITY, COURANT INSTITUTE — New York, NY

2013/09–2014/10 Visiting Scholar; advisor: Jinyang Li

Education

TSINGHUA UNIVERSITY — Beijing, China

Ph.D. in Computer Science, 2015

Advisors: Yongwei Wu and Kang Chen; supervisor: Weimin Zheng

CHINA AGRICULTURAL UNIVERSITY — Beijing, China

B.S. in Computer Science, 2010

Selected Honors

ACM SIGMOD Research Highlight Award, *CausalMesh*.

Stony Brook University Research Excellence Award, 2024.

NSF CAREER Award, 2023.

Committee, Panel, and Referee Service

2027 NSDI, OSDI (upcoming program committees)

2026 OSDI, ASPLOS, EuroSys, USENIX ATC

2025 EuroSys, USENIX ATC

2024 SOSP, USENIX ATC

2023 NSDI, ICDCS, SIGCOMM

2022 ASPLOS (external review committee)

2021 ApSys, NSDI, IEEE TPDS

2020 ApSys, USENIX ATC, *Distributed Computing*, NSF CNS Small, IEEE TPDS

2019 ApSys, *Communications of the ACM*

2018 IEEE TPDS

Publications and Accepted Papers

[1] Ti Zhou, Zihao Zhang, Omar Chowdhury, and Shuai Mu. **Lion: Modular Verification of Async Runtime Liveness**. ACM Symposium on Operating Systems Principles (SOSP), 2026; accepted.

[2] Ze Tang, Zihao Zhang, Weihai Shen, Jicheng Shi, and Shuai Mu. **Jetpack: Consensus Made Generally Fast**. USENIX Symposium on Operating Systems Design and Implementation (OSDI), 2026.

[3] Zihao Zhang, Ti Zhou, Christa Jenkins, Omar Chowdhury, and Shuai Mu. **AutoMan: Facilitating Verified Distributed Systems Development Through Automatic Code Generation and Manual Optimizations**. ACM Symposium on Operating Systems Principles (SOSP), 2025.

[4] Jinkun Geng, Shuai Mu, Anirudh Sivaraman, and Balaji Prabhakar. **Tiga: Accelerating Geo-Distributed Transactions with Synchronized Clocks**. ACM Symposium on Operating Systems Principles (SOSP), 2025.

[5] Weihai Shen, Yang Cui, Siddhartha Sen, Sebastian Angel, and Shuai Mu. **Mako: Speculative Distributed Transactions with Geo-Replication**. USENIX Symposium on Operating Systems Design and Implementation (OSDI), 2025.

[6] Haoran Zhang, Shuai Mu, Sebastian Angel, and Vincent Liu. **CausalMesh: A Causal Cache for Stateful Serverless Computing**. *SIGMOD Record*, 54(1):48–57, 2025. Research Highlight article.

[7] Haoran Zhang, Shuai Mu, Sebastian Angel, and Vincent Liu. **CausalMesh: A Causal Cache for Stateful Serverless Computing**. *Proceedings of the VLDB Endowment*, 17(13), 2024; presented at VLDB 2025. Best Paper Award nominee.

[8] Haonan Lu, Shuai Mu, Wyatt Lloyd, and Siddhartha Sen. **NCC: Natural Concurrency Control for Strictly Serializable Datastores by Avoiding the Timestamp-Inversion Pitfall**. USENIX Symposium on Operating Systems Design and Implementation (OSDI), 2023.

[9] Jian Zhang, Ye Ji, Shuai Mu, and Cheng Tan. **Viper: A Fast Snapshot Isolation Checker**. European Conference on Computer Systems (EuroSys), 2023.

[10] Mohammadreza Alimadadi, Hieu Mai, Shenghsun Cho, Michael Ferdman, Peter Milder, and Shuai Mu. **Waverunner: An Elegant Approach to Hardware Acceleration of State Machine Replication**. USENIX Symposium on Networked Systems Design and Implementation (NSDI), 2023.

[11] Xuhao Luo, Weihai Shen, Shuai Mu, and Tianyin Xu. **DepFast: Orchestrating Code of Quorum Systems**. USENIX Annual Technical Conference (ATC), 2022.

[12] Weihai Shen, Ansh Khanna, Sebastian Angel, Siddhartha Sen, and Shuai Mu. **Rolis: A Software Approach to Efficiently Replicating Multi-Core Transactions**. European Conference on Computer Systems (EuroSys), 2022.

Publications (continued)

[13] Andrew Yoo, Yuanli Wang, Ritesh Sinha, Shuai Mu, and Tianyin Xu. **Fail-Slow Fault Tolerance Needs Programming Support**. Workshop on Hot Topics in Operating Systems (HotOS), 2021.

[14] Siyuan Zhou and Shuai Mu. **Fault-Tolerant Replication with Pull-Based Consensus in MongoDB**. USENIX Symposium on Networked Systems Design and Implementation (NSDI), 2021.

[15] Cheng Tan, Changgeng Zhao, Shuai Mu, and Michael Walfish. **Cobra: Making Transactional Key-Value Stores Verifiably Serializable**. USENIX Symposium on Operating Systems Design and Implementation (OSDI), 2020.

[16] Zhaoguo Wang, Changgeng Zhao, Shuai Mu, Haibo Chen, and Jinyang Li. **On the Parallels between Paxos and Raft, and How to Port Optimizations**. ACM Symposium on Principles of Distributed Computing (PODC), 2019.

[17] Shuai Mu, Sebastian Angel, and Dennis Shasha. **Deferred Runtime Pipelining for Contentious Multicore Software Transactions**. European Conference on Computer Systems (EuroSys), 2019.

[18] Yu Lin Chen, Shuai Mu, Jinyang Li, Cheng Huang, Jin Li, Aaron Ogus, and Douglas Phillips. **Giza: Erasure Coding Objects across Global Data Centers**. USENIX Annual Technical Conference (ATC), 2017.

[19] Haonan Lu, Christopher Hodsdon, Khiem Ngo, Shuai Mu, and Wyatt Lloyd. **The SNOW Theorem and Latency-Optimal Read-Only Transactions**. USENIX Symposium on Operating Systems Design and Implementation (OSDI), 2016.

[20] Shuai Mu, Lamont Nelson, Wyatt Lloyd, and Jinyang Li. **Consolidating Concurrency Control and Consensus for Commits under Conflicts**. USENIX Symposium on Operating Systems Design and Implementation (OSDI), 2016.

[21] Zhaoguo Wang, Shuai Mu, Yang Cui, Han Yi, Haibo Chen, and Jinyang Li. **Scaling Multicore Databases via Constrained Parallel Execution**. ACM SIGMOD International Conference on Management of Data, 2016.

[22] Shuai Mu, Yang Cui, Yang Zhang, Wyatt Lloyd, and Jinyang Li. **Extracting More Concurrency from Distributed Transactions**. USENIX Symposium on Operating Systems Design and Implementation (OSDI), 2014.

[23] Shuai Mu, Kang Chen, Yongwei Wu, and Weimin Zheng. **When Paxos Meets Erasure Code: Reduce Network and Storage Cost in State Machine Replication**. International Symposium on High-Performance Parallel and Distributed Computing (HPDC), 2014.

[24] Shuai Mu, Kang Chen, Pin Gao, Feng Ye, Yongwei Wu, and Weimin Zheng. **μLibCloud: Providing High Available and Uniform Accessing to Multiple Cloud Storages**. ACM/IEEE International Conference on Grid Computing, 2012.

Teaching

Stony Brook University

CSE 532: Database Systems — Fall 2025, Fall 2026.
CSE 590: Reliable System Programming — Spring 2025, Spring 2026.
CSE 535: Distributed Systems — Spring 2019; Fall 2019, 2020, 2022, 2023.
CSE 416: Software Engineering — Spring 2022, Spring 2024; Fall 2024, 2025, 2026.
CSE 643: Seminar in Concurrency — Fall 2020; Spring 2021; Fall 2023; Spring and Fall 2024.
Supervision of undergraduate research, graduate advanced projects, independent study, master's theses, doctoral dissertations, and teaching practica.

New York University

CSCI-UA.0201: Computer Systems Organization — Spring 2018.

Advising and Mentoring

Current research group

Zihao Zhang — Postdoctoral researcher.
Jinkun Geng — Postdoctoral researcher, co-advised with Anirudh Sivaraman.
Ze Tang, Zeyu Yang, Ti Zhou — Ph.D. students.

Selected alumni

Weihai Shen — Ph.D., 2026; Google.
Krish Makadia — M.S. thesis.
Yida Wu — M.S. thesis; Cloudera.
Chenyu Wang — M.S. advanced project; Meta.
Mrityunjay Kumar — M.S. advanced project; VMware.
Satya Jain — M.S. advanced project; VMware.
Ritesh Sinha — M.S. advanced project; HP.
Ansh Khanna — M.S. advanced project; Google.
Julie Lee — B.S./M.S.; HP.
Ayush Jain — M.S.; Oracle.
Devika Sudheer — M.S. advanced project; LinkedIn.
Radhika Agarwal — M.S. advanced project; Apple.
Nilesh Gajwani, Ray Zhang, Sethu Karthikeyan, Saumitra Bose — M.S. advanced projects.

Collaborative mentoring

Jian Zhang — Ph.D., Northeastern University, advised by Cheng Tan; Google.
Haoran Zhang — Ph.D., University of Pennsylvania, advised by Vincent Liu and Sebastian Angel; Meta.
Andrew Yoo — M.S. thesis, University of Illinois Urbana-Champaign, with Tianyin Xu; Whova.
Yuanli Wang — Virtual intern; M.S., University of Minnesota; Ph.D. student, Boston University.